

PE-EC701A — Microwave Theory and Technique

Semester VII • 3L:0T:0P • 3 Credits

Module 6 Exam Notes: Microwave Design Principles

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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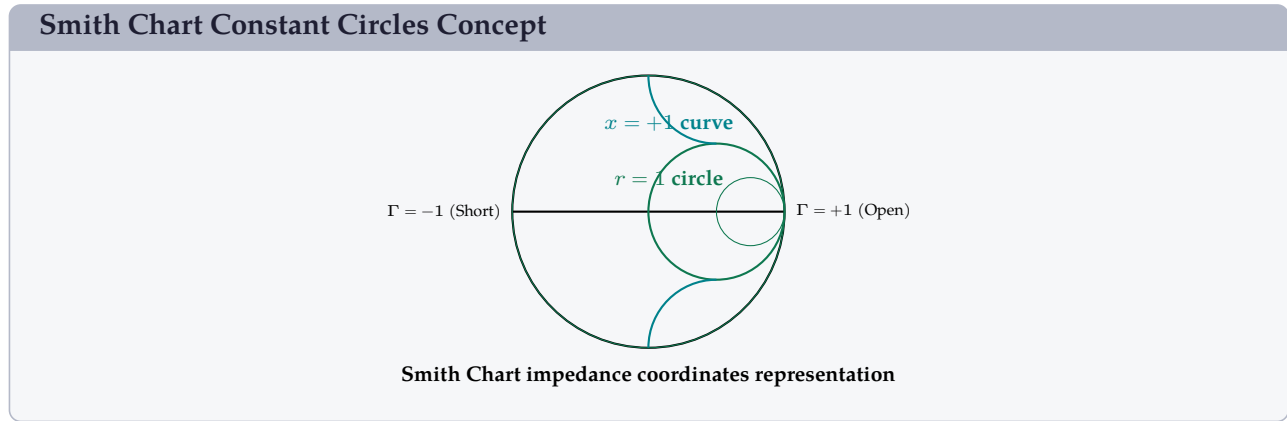
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1. Impedance Transformation and Matching

Impedance matching is crucial to ensure maximum power transfer and prevent reflections in microwave networks.

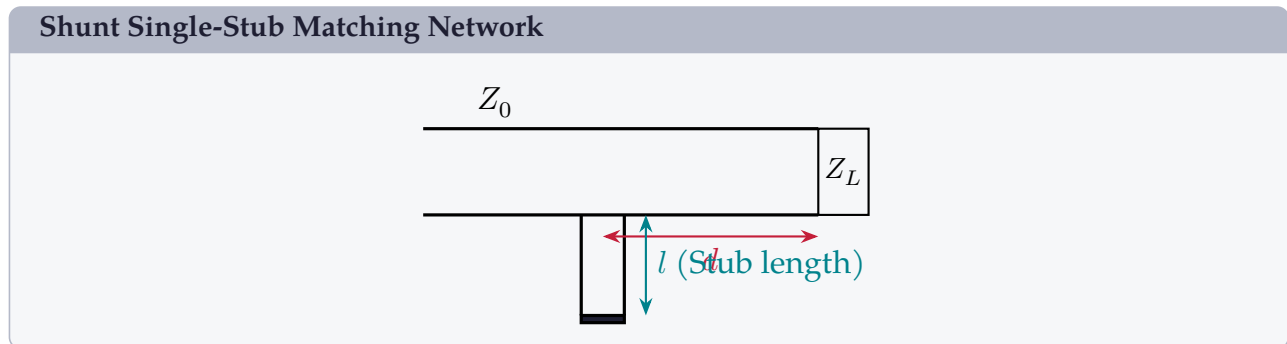
1.1 Smith Chart Coordinates

The Smith Chart is a polar plot of the complex reflection coefficient (Γ), mapped to constant resistance (r) and constant reactance (x) circles:



1.2 Single-Stub Matching

A single-stub tuner uses a shunt (or series) short- or open-circuited transmission line stub placed at a distance d from the load to match a load impedance Z_L to the line impedance Z_0 .



1.2.1 Design Equations for Shunt Single-Stub

Let $Y_L = 1/Z_L = G_L + jB_L$ be the load admittance. We normalize to $Y_0 = 1/Z_0$:

$$y_L = g_L + jb_L = \frac{Y_L}{Y_0} \quad (1)$$

The distance d is chosen such that the admittance looking into the line at d is:

$$y_d = 1 + jb \quad (2)$$

The stub length l is then selected to provide a susceptance of $-jb$, canceling the reactive part and leaving a perfect match ($y_{in} = y_d + y_{stub} = 1$).

$$d = \frac{\lambda}{2\pi} \tan^{-1} \left(\frac{r_L - 1}{x_L} \right) \quad (\text{simplified for specific cases}) \quad (3)$$

$$l_s = -\frac{\lambda}{2\pi} \tan^{-1}(b) \quad (\text{short-circuited stub}) \quad (4)$$

2. Microwave Filter Design

Filters restrict frequencies allowed to pass. The modern **Insertion Loss Method** uses low-pass filter (LPF) prototypes with normalized values (g_i):

- **Maximally Flat (Butterworth):** Flat passband response, roll-off is smooth.
- **Equal Ripple (Chebyshev):** Sharp cutoff, containing ripples in the passband.

Butterworth Prototype Values

For a maximally flat low-pass filter, the element values g_i ($1 \leq i \leq N$) are:

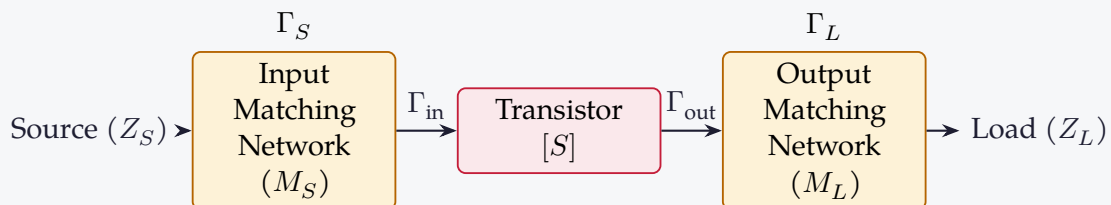
$$g_i = 2 \sin \left[\frac{(2i-1)\pi}{2N} \right] \quad \text{for } i = 1, 2, \dots, N \quad (5)$$

And $g_{N+1} = 1.0$ for all N .

3. RF and Microwave Amplifier Design

Microwave amplifier design utilizes active devices (transistors) characterized by their S-parameters:

Microwave Transistor Amplifier Block Diagram



3.1 Amplifier Stability Criteria

An amplifier is **unconditionally stable** if the transistor will not oscillate for any passive source and load impedance ($\text{Re}(Z_S), \text{Re}(Z_L) > 0$). This requires the Rollett stability factor (K) to exceed 1 and the matrix determinant delta ($|\Delta|$) to be less than 1:

Rollett Stability Criteria

$$K = \frac{1 - |S_{11}|^2 - |S_{22}|^2 + |\Delta|^2}{2|S_{12}S_{21}|} > 1 \quad (6)$$

$$|\Delta| = |S_{11}S_{22} - S_{12}S_{21}| < 1 \quad (7)$$

If $K < 1$, the amplifier is only **conditionally stable**, and matching networks (Γ_S, Γ_L) must be chosen outside the unstable regions defined by the **stability circles**.

3.2 Amplifier Power Gains

- **Transducer Power Gain (G_T):** The ratio of power delivered to the load to the power available from the source.

$$G_T = \frac{1 - |\Gamma_S|^2}{|1 - S_{11}\Gamma_S|^2} |S_{21}|^2 \frac{1 - |\Gamma_L|^2}{|1 - S_{22}\Gamma_L - \frac{S_{12}S_{21}\Gamma_S\Gamma_L}{1 - S_{11}\Gamma_S}|^2} \quad (8)$$

- **Unilateral Transducer Gain (G_{TU}):** Assumes negligible feedback ($S_{12} = 0$).

$$G_{TU} = \frac{1 - |\Gamma_S|^2}{|1 - S_{11}\Gamma_S|^2} |S_{21}|^2 \frac{1 - |\Gamma_L|^2}{|1 - S_{22}\Gamma_L|^2} \quad (9)$$

3.3 Low Noise Amplifier (LNA) Design & Noise Figure Circles

In LNA design (critical for receiver front-ends), the goal is to achieve the lowest possible Noise Figure (F), even at the expense of maximum transducer gain.

- **Noise Figure Formula:**

$$F = F_{\min} + \frac{4R_n}{Z_0} \frac{|\Gamma_S - \Gamma_{\text{opt}}|^2}{(1 - |\Gamma_S|^2)|1 + \Gamma_{\text{opt}}|^2} \quad (10)$$

where $F_{\min}$ is the minimum noise figure of the transistor, R_n is the equivalent noise resistance, Γ_{opt} is the optimum source reflection coefficient, and Γ_S is the actual source reflection coefficient.

- **Noise Figure Circles:** For a target noise figure $F > F_{\min}$, the locus of all source reflection coefficients Γ_S forms a circle on the Smith Chart.

- **Center (C_F):**

$$C_F = \frac{\Gamma_{\text{opt}}}{1 + N} \quad (11)$$

- **Radius (R_F):**

$$R_F = \frac{\sqrt{N^2 + N(1 - |\Gamma_{\text{opt}}|^2)}}{1 + N} \quad (12)$$

- **Noise Parameter (N):**

$$N = \frac{F - F_{\min}}{4R_n/Z_0} |1 + \Gamma_{\text{opt}}|^2 \quad (13)$$

The designer selects a source reflection coefficient Γ_S lying on the desired Noise Figure circle that offers a reasonable trade-off with input matching and gain.

4. Microwave Mixer Design

Mixers are 3-port non-linear devices that perform frequency translation by multiplying the RF input signal (f_{RF}) with a Local Oscillator signal (f_{LO}), producing an Intermediate Frequency ($f_{\text{IF}} = |f_{\text{RF}} \pm f_{\text{LO}}|$).

- **Single-Ended Diode Mixer:** Uses a single Schottky diode. It is simple but provides poor isolation between the RF and LO ports, and does not suppress LO amplitude noise.
- **Balanced Mixer:** Utilizes two Schottky diodes combined via a 3 dB hybrid junction (such as a Magic Tee or branch-line coupler).
 - **LO Noise Suppression:** The hybrid junction feeds the LO signal to the two diodes out-of-phase (180°) while feeding the RF signal in-phase. The combined output at the IF port cancels out the LO amplitude noise.
 - **Isolation:** Provides high isolation (> 20 dB) between the RF and LO ports, preventing local oscillator power leakage back into the antenna.
- **Conversion Loss (L_C):** The ratio of RF input power to the desired IF output power:

$$L_C = 10 \log_{10} \left(\frac{P_{\text{RF}}}{P_{\text{IF}}} \right) \quad (\text{dB}) \quad (14)$$

For passive diode mixers, typical conversion loss ranges from 5 dB to 8 dB.

5. Microwave Oscillator Design

Microwave oscillators convert DC power into RF power using active devices (transistors with positive feedback, or negative resistance diodes like Gunn and IMPATT).

5.1 One-Port Negative Resistance Oscillator

An active device presenting a negative input impedance $Z_{\text{in}} = R_{\text{in}} + jX_{\text{in}}$ (where $R_{\text{in}} < 0$) is terminated with a passive load impedance $Z_L = R_L + jX_L$:

- **Startup Condition:** For oscillations to initiate from thermal noise:

$$R_{\text{in}} + R_L < 0 \quad \text{and} \quad X_{\text{in}} + X_L = 0 \quad (15)$$

- **Steady-State Stability:** As the oscillation amplitude A builds up, the negative resistance $R_{\text{in}}(A)$ saturates (becomes less negative) due to device non-linearities until a stable operating amplitude A_0 is reached:

$$R_{\text{in}}(A_0) + R_L = 0 \quad \text{and} \quad X_{\text{in}}(A_0) + X_L = 0 \quad (16)$$

5.2 Two-Port Transistor Oscillator

A transistor-based oscillator uses an active three-terminal device (e.g., GaAs MESFET). An unstable termination at one port (such as a capacitive load at the source) is used to generate a negative input resistance at the other port (gate), which is then connected to a high- Q resonant circuit (tuning network) to establish oscillation at the desired frequency.

5.3 Microwave Power Amplifier Design

Microwave power amplifiers (PAs) are designed to deliver high RF power levels to a load (typically a transmitting antenna) with maximum efficiency and minimal distortion.

- **1-dB Compression Point ($P_{1\text{dB}}$):** As input power increases, the amplifier eventually saturates and gain drops. The $P_{1\text{dB}}$ is the power level at which the output power is 1 dB lower than it would be in the ideal linear region. It marks the boundary of the linear operating region.
- **Power Added Efficiency (PAE):** A critical metric for PAs that accounts for the input RF power, expressing the net RF power generated relative to the DC power consumed:

$$\text{PAE} = \frac{P_{\text{RF, out}} - P_{\text{RF, in}}}{P_{\text{DC}}} \times 100\% \quad (17)$$

- **Drain/Collector Efficiency (η):** Ratio of output RF power to input DC power: $\eta = P_{\text{RF, out}}/P_{\text{DC}} \times 100\%$.

Power Amplifier Classes:

PAs are classified based on their conduction angle (θ_c), which defines the fraction of the RF cycle during which the active transistor conducts current.

Class	Conduction Angle (θ_c)	Ideal Max Efficiency	Linearity	Characteristics
Class A	360° (conducts full cycle)	50%	Outstanding	Low efficiency; high heat dissipation; active at all times.
Class AB	180° < θ_c < 360°	50% – 78.5%	Good	Compromise between Class A and B; commonly used in RF links.
Class B	180° (conducts half cycle)	78.5%	Moderate	Transistor conducts only during one half-cycle; requires push-pull.
Class C	< 180° (conducts narrow pulses)	> 85% (approaching 100%)	Very Poor	High distortion; requires resonant tank circuit to reconstruct wave.

Quick Revision — Module VI

Microwave Design Principles | PE-EC701A

Key Formulas at a Glance

Rollett Stability factor	$K = \frac{1- S_{11} ^2- S_{22} ^2+ \Delta ^2}{2 S_{12}S_{21} }$
Determinant delta	$\Delta = S_{11}S_{22} - S_{12}S_{21}$
Unilateral gain	$G_{TU} = \frac{1- \Gamma_S ^2}{ 1-S_{11}\Gamma_S ^2} S_{21} ^2 \frac{1- \Gamma_L ^2}{ 1-S_{22}\Gamma_L ^2}$
Butterworth filter	$g_i = 2 \sin \left(\frac{(2i-1)\pi}{2N} \right)$
Unilateral figure merit	$U = \frac{ S_{11}S_{12}S_{21}S_{22} }{(1- S_{11} ^2)(1- S_{22} ^2)}$
LNA Noise Parameter (N)	$N = \frac{F-F_{\min}}{4R_n/Z_0} 1 + \Gamma_{\text{opt}} ^2$
LNA Circle Center (C_F)	$C_F = \frac{\Gamma_{\text{opt}}}{1+N}$
LNA Circle Radius (R_F)	$R_F = \frac{\sqrt{N^2+N(1- \Gamma_{\text{opt}} ^2)}}{1+N}$

One-Line Definitions

- **Smith Chart:** A graphical utility mapping reflection coefficients to complex impedances.
- **Stub Tuning:** Impedance matching using shunt or series transmission line segments.
- **Butterworth Filter:** A filter designed for a maximally flat amplitude response in the passband.
- **Chebyshev Filter:** A filter that provides sharp roll-off at the expense of equal-ripple passband ripples.
- **Rollett Stability:** Mathematical check ($K > 1$) determining if an amplifier is unconditionally stable.
- **LNA:** Low Noise Amplifier, designed to minimize noise figure circles on the input matching network.
- **Stability Circles:** Regions plotted on the Smith Chart enclosing unstable source/load reflection coefficients.
- **Conversion Loss:** The ratio of RF input power to IF output power in a passive mixer (expressed in dB).
- **Balanced Mixer:** A mixer using a hybrid junction to combine two diodes, canceling LO noise and providing port isolation.
- **One-Port Oscillator:** An oscillator designed by terminating a negative-resistance device with a resonant load.

Frequently Asked / Viva Questions

Q1. What are constant resistance and constant reactance circles on a Smith Chart?

Constant resistance circles are circles whose centers lie on the horizontal real axis, all passing through the open-circuit point. Constant reactance circles are arcs representing imaginary parts.

Frequently Asked / Viva Questions (contd.)**Q2. What is the difference between single-stub and double-stub matching?**

Single-stub matching requires adjusting both the stub length and its position from the load. Double-stub matching uses fixed-position stubs, adjusting only their lengths.

Q3. State the conditions for unconditional stability in microwave amplifiers.

The Rollett stability factor K must be greater than 1 ($K > 1$) and the S-parameter determinant magnitude $|\Delta|$ must be less than 1 ($|\Delta| < 1$).

Q4. Why are short-circuited stubs preferred over open-circuited stubs?

Because open-circuited stubs radiate energy from their open ends, behaving as antennas. Short-circuited stubs confine fields and are easier to manufacture with high precision.

Q5. What is the unilateral assumption in amplifier design?

The assumption that $S_{12} = 0$, meaning there is no feedback from output to input. This simplifies the design of matching networks.

Q6. Explain the insertion loss method of filter design.

It designs filters based on a power transfer function (like Butterworth or Chebyshev), yielding element values (g_i) normalized to a $1\ \Omega$ source and 1 rad/s cutoff.

Q7. What is the significance of Noise Figure Circles in LNA design?

They map the source reflection coefficients (Γ_S) that produce a constant noise figure, allowing the designer to balance noise figure and gain.

Q8. Under what condition is single-stub matching not possible?

When the load impedance lies within a specific prohibited region of the Smith Chart (for a given shunt or series configuration), requiring a different stub location.

Q9. What are Richards' Transformations?

Mathematical maps that convert lumped inductors and capacitors into transmission line stubs ($L \rightarrow$ short stub, $C \rightarrow$ open stub) at microwave frequencies.

Q10. What is Kuroda's Identity?

A set of network equivalence rules used to separate filter stubs using redundant transmission line sections (unit elements).

Q11. Explain the difference between power gain, available gain, and transducer gain.

Transducer gain accounts for mismatch at both input and output. Available gain assumes matched output. Power gain assumes matched input.

Frequently Asked / Viva Questions (contd.)

Q12. What is the Unilateral Figure of Merit (U)?

A ratio used to calculate the maximum error introduced when assuming $S_{12} = 0$ instead of using the bilateral gain equations.

Q13. State the startup and steady-state conditions for a negative resistance one-port oscillator.

Startup: $R_{in} + R_L < 0$ and $X_{in} + X_L = 0$ (net loop resistance is negative, allowing oscillations to build from thermal noise). Steady-state: $R_{in}(A_0) + R_L = 0$ and $X_{in}(A_0) + X_L = 0$ (saturation reduces negative resistance until net loop resistance is zero, sustaining a stable amplitude A_0).

Q14. How does a balanced mixer achieve LO noise suppression?

By using a 3 dB hybrid junction (such as a Magic Tee) that feeds the local oscillator (LO) signal to the two diodes 180° out-of-phase, while feeding the RF signal in-phase. The combined IF output cancels the LO amplitude noise components.

Comparison of Single-Stub and Double-Stub Tuning

Feature	Single-Stub Tuning	Double-Stub Tuning
Stub Position	Must be variable (requires sliding stub or custom line length).	Fixed spacing (typically $\lambda/8$ or $3\lambda/8$).
Adjustment	Adjusts both stub length and stub distance from the load.	Adjusts only the lengths of the two stubs.
Matching Range	Can match any load impedance with non-zero real part.	Cannot match all loads (contains a "forbidden region").
Implementation	Difficult in microstrip (requires changing physical layout).	Easy in microstrip (stubs are printed at fixed locations).